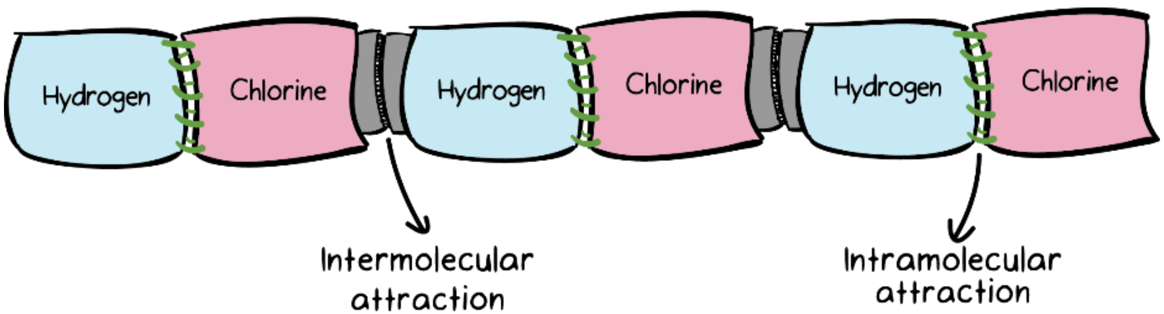
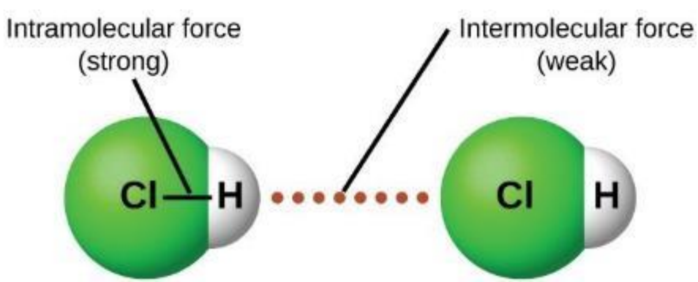
Module 4

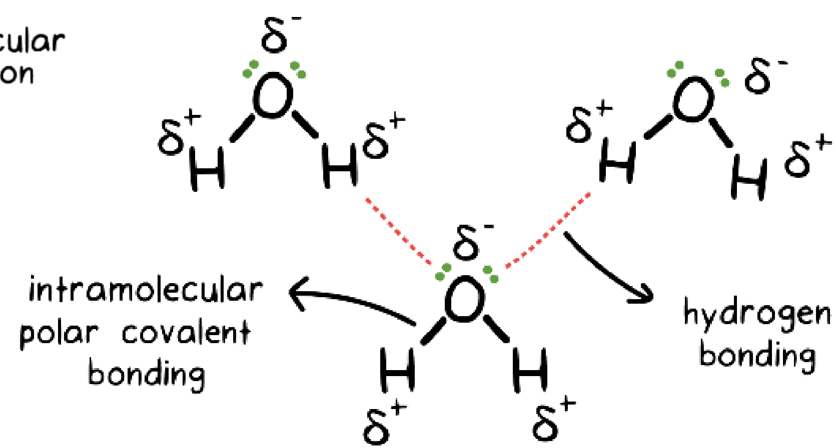
Potential Energy Surfaces

Molecular Forces

- **Intermolecular forces** are attractive forces **between** molecules.
- **Intramolecular forces** hold atoms together in a molecule.

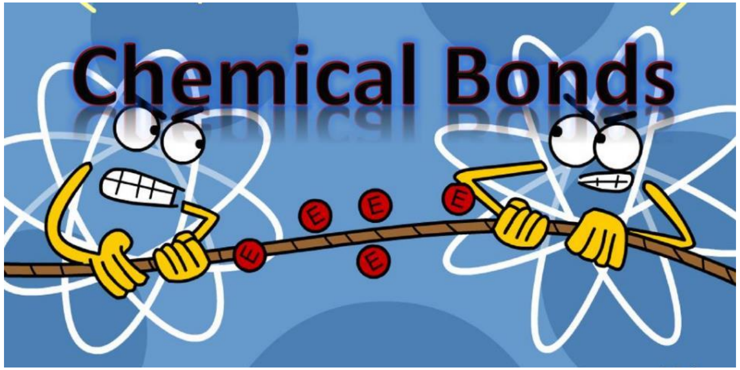




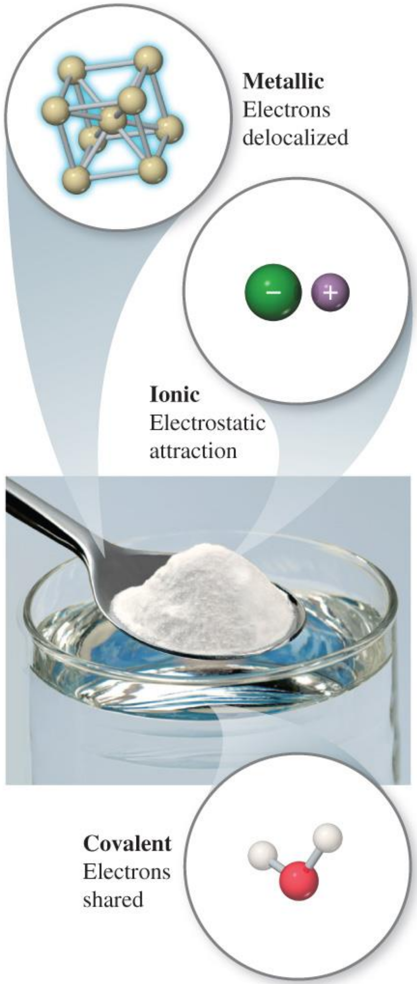


Chemical Bonding

- A chemical bond is an interaction between atoms, ions or molecules that enables the formation of chemical compounds.
- It is a mutual attraction between the nuclei and valence electrons of different atoms that bonds atom together.



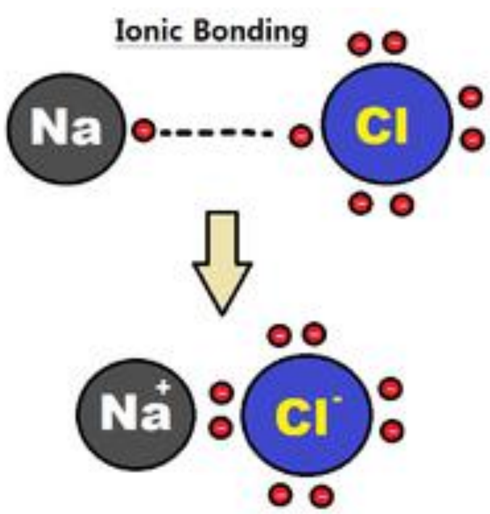
Chemical Bonds

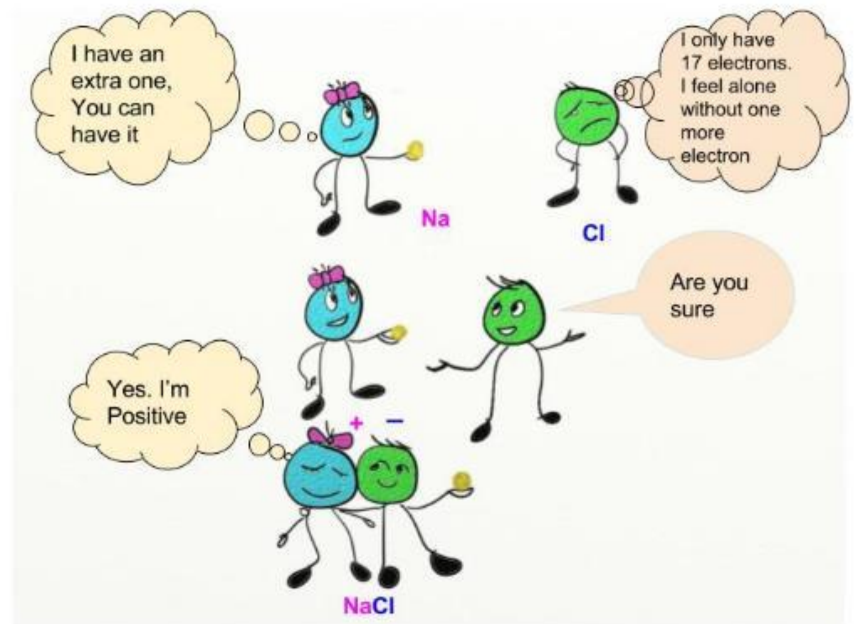


- Three basic types of bonds
  - Ionic
    - Electrostatic attraction between ions.
  - Covalent
    - Sharing of electrons.
  - Metallic
    - Metal atoms bonded to several other atoms.

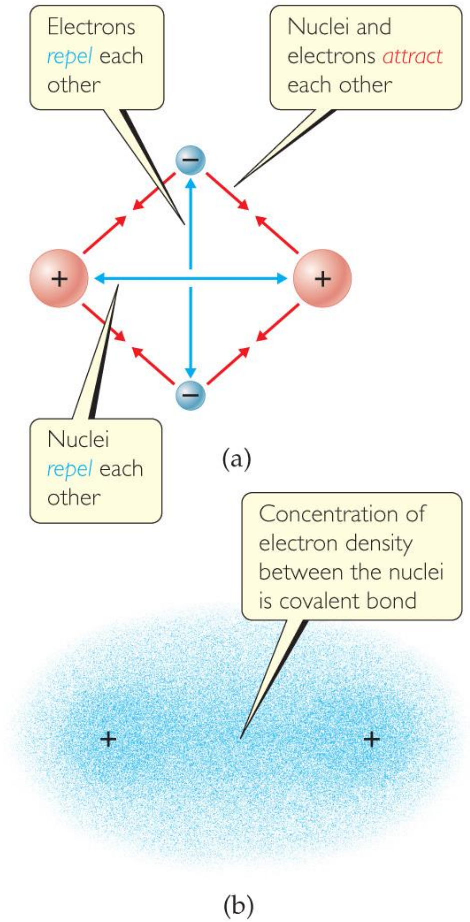
Ionic bond

- Electrons are transferred from one atom to another
- Occurs between one metal and one non-metal
- Metals loose electrons to become cations
- Non-metals gain electrons to become anions
- Examples: NaCl, $MgSO_4$





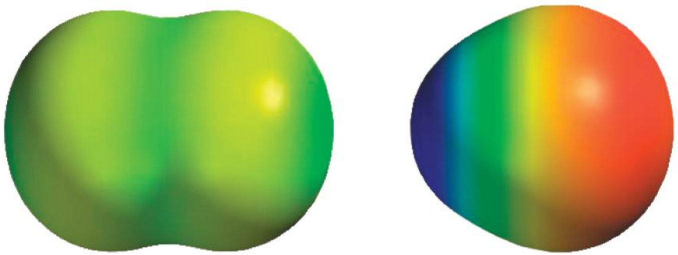
Covalent Bonding



- In covalent bonds, atoms share electrons.
- There are several electrostatic interactions in these bonds:
  - ➢ Attractions between electrons and nuclei,
  - ➢ Repulsions between electrons,
  - ➢ Repulsions between nuclei.

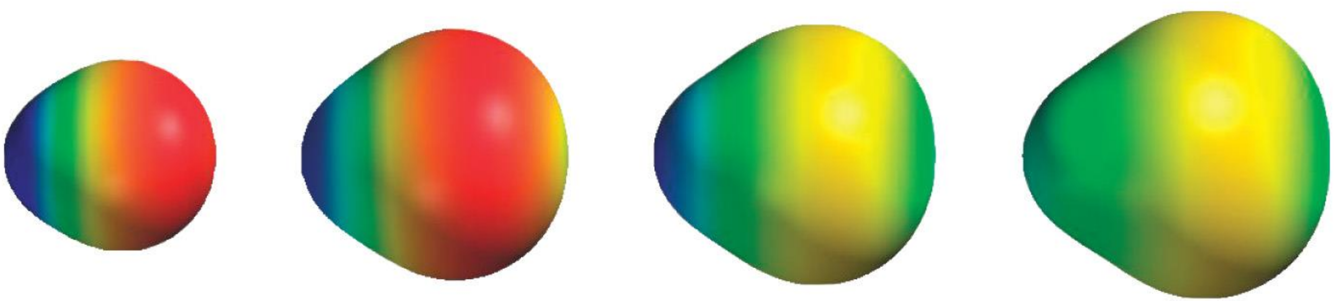
Bond energies range from 150-700 kJ/mol

Polar Covalent Bonds



F_2 HF

- Though atoms often form compounds by sharing electrons, the electrons are not always shared equally.
- Fluorine pulls harder on the electrons it shares with hydrogen than hydrogen does.
- Therefore, the fluorine end of the molecule has more electron density than the hydrogen end.
- The greater the difference in electronegativity, the more polar is the bond.



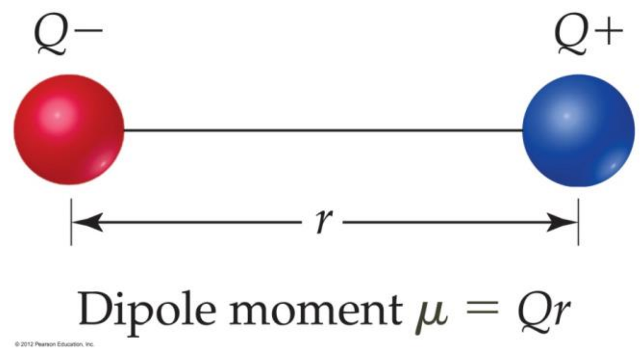
HF HCl HBr HI

Polar Covalent Bonds

- When two atoms share electrons unequally, a **bond dipole** results.
- The **dipole moment**, μ, produced by two equal but opposite charges separated by a distance, r, is calculated:

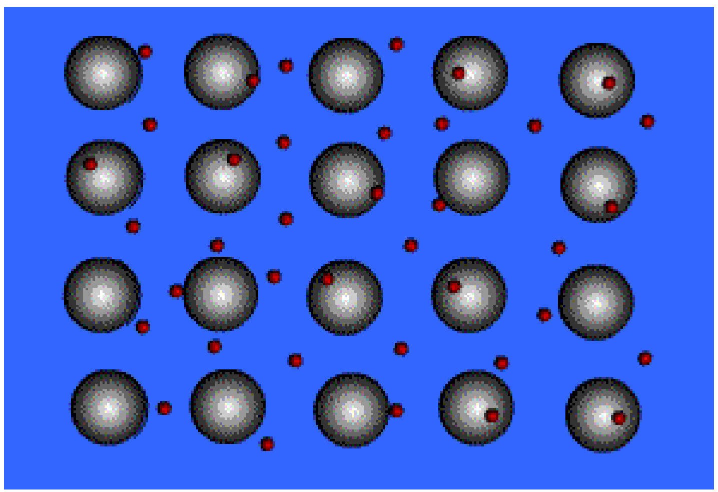
$$\mu = Qr$$

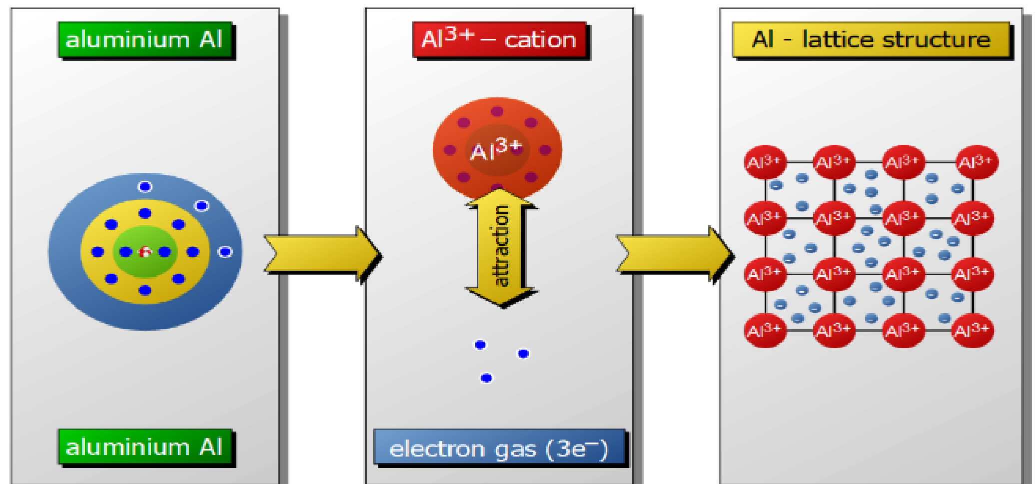
- It is measured in debyes (D).
- 1 D = 3.33564 × 10^{-30} C.m, where C is Coulomb and m denotes a meter.



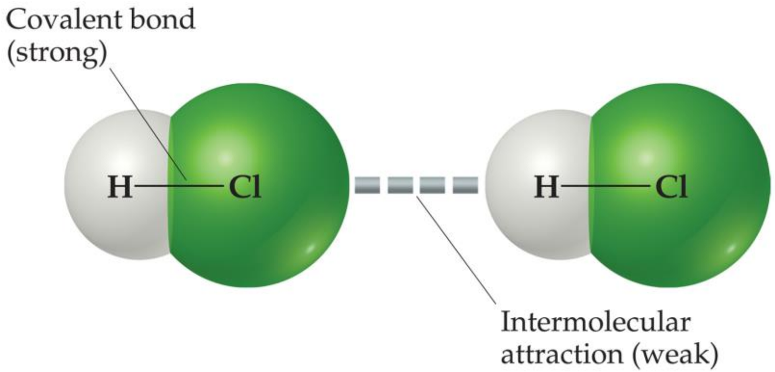
Metallic bond

- Metal atoms share many electrons in a ‘sea’ that is free to move throughout the metal.
- Metallic bond results from attractions between the metal cations and the surrounding sea of electrons
- Vacant p and d orbitals in metal’s outer energy levels overlap, and allow outer electrons to move freely throughout the metal.
- Valence electrons do not belong to any one atom
- Examples: metals and alloys





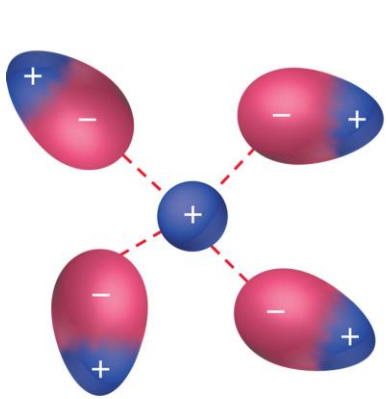
Intermolecular Forces



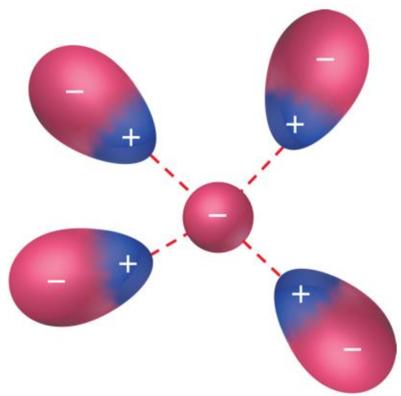
- The attractions between molecules are not nearly as strong as the intramolecular attractions that hold compounds together.
- They are, however, strong enough to control physical properties such as boiling and melting points, vapor pressures, and viscosities.

Ion-Dipole interactions

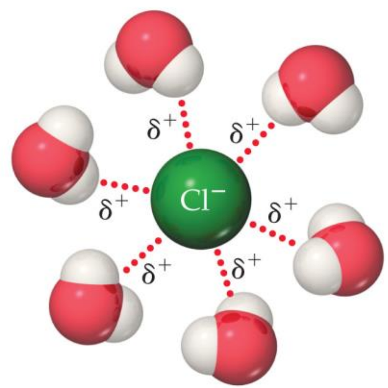
- An ion-dipole force exists between an ion and the partial charge on the end of a polar molecule/ neutral molecule with a dipole.
- The strength of these forces are what make it possible for ionic substances to dissolve in polar solvents.
- Ion-dipole attractions become strong as the charge on the ion increases or the magnitude of dipole on the polar molecule increases.



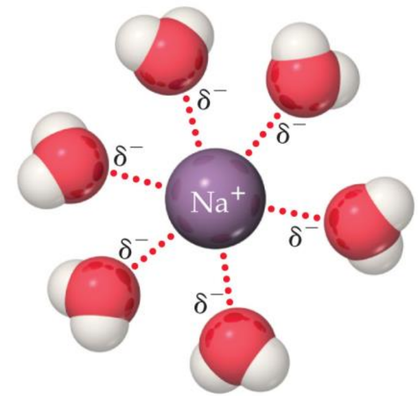
Cation-dipole attractions



Anion-dipole attractions



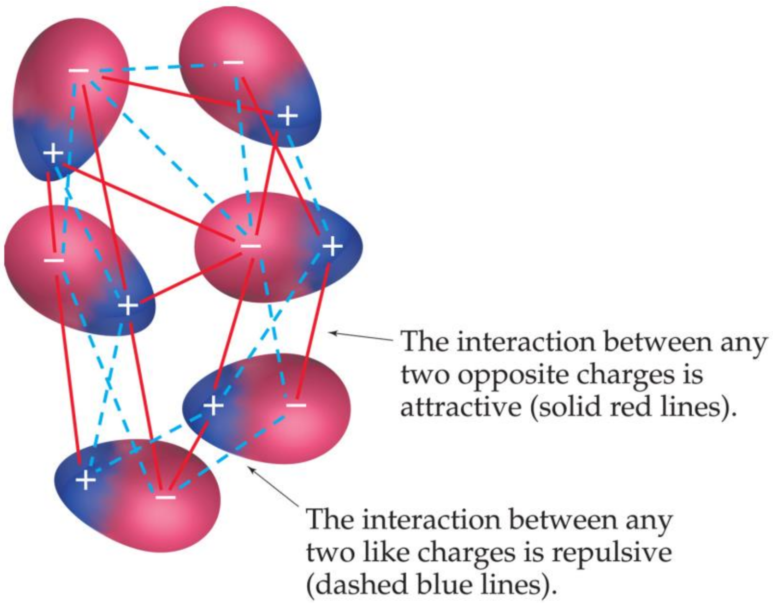
Positive ends of polar molecules are oriented toward negatively charged anion

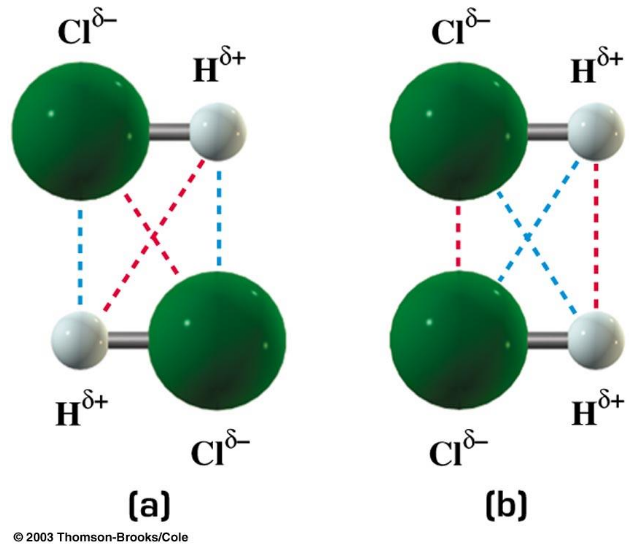


Negative ends of polar molecules are oriented toward positively charged cation

Dipole-Dipole interactions

- Molecules that have permanent dipoles are attracted to each other.
  - The positive end of one is attracted to the negative end of the other and vice-versa.
  - These forces are only important when the molecules are close to each other.





Dipole-Dipole interactions

Substance	Molecular Weight (amu)	Dipole Moment μ (D)	Boiling Point (K)
Propane, $CH_3CH_2CH_3$	44	0.1	231
Dimethyl ether, CH_3OCH_3	46	1.3	248
Methyl chloride, CH_3Cl	50	1.9	249
Acetaldehyde, CH_3CHO	44	2.7	294
Acetonitrile, CH_3CN	41	3.9	355

The more polar the molecule, the higher is its boiling point.

Directional forces

- Which forces require molecules to be aligned such that the interactions are in optimal direction?

Ion-dipole and dipole-dipole forces

→ ***directional*** **in nature**

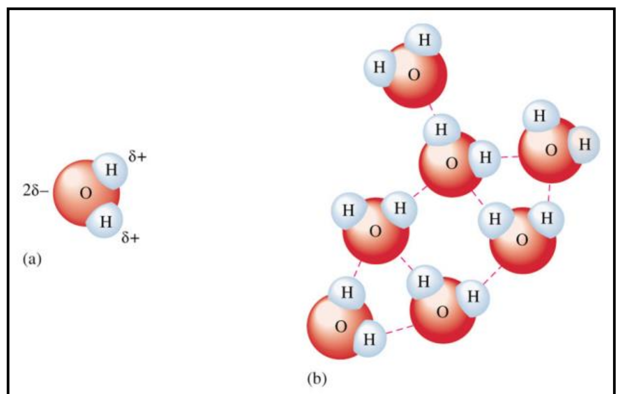
- Which forces do not require molecules to be aligned in a particular direction?

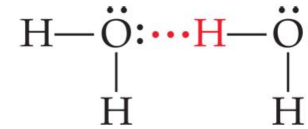
Ion-ion interactions

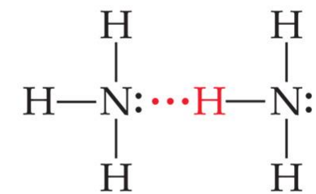
→ ***non-directional*** **in nature**

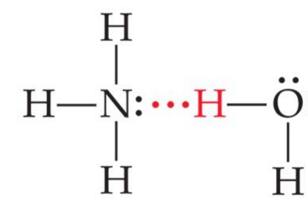
Dipole-Dipole Interactions: Hydrogen Bonding

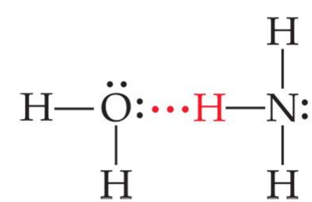
- Special type of dipole-dipole attraction between molecules.
- When H bonded to a highly electronegative atom (such as N, O, F), the bonding electron pair is drawn towards the electronegative atom.
- NOT a covalent bond to a H atom.
- The energies of H bonds range 4 to 25 kJ/mol. Regular covalent bond strength range between 150 and 1100 kJ/mol.





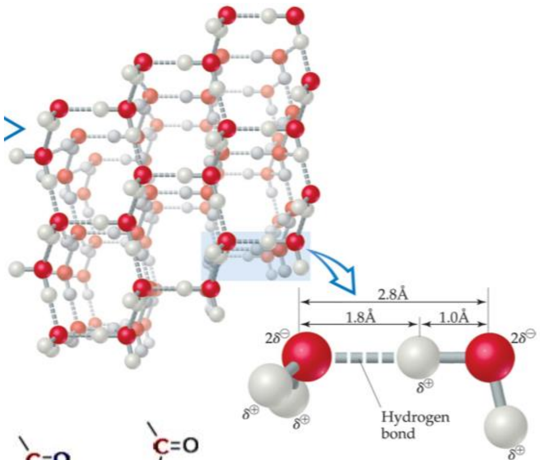


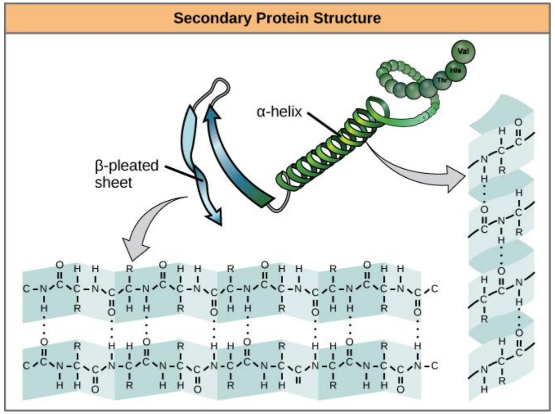


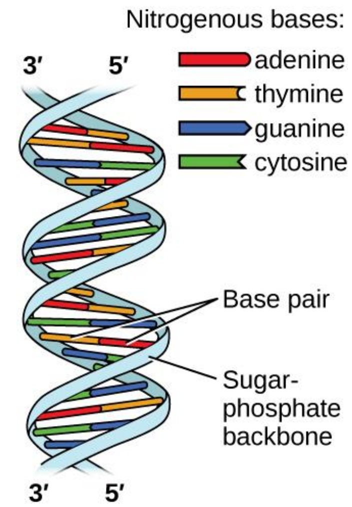


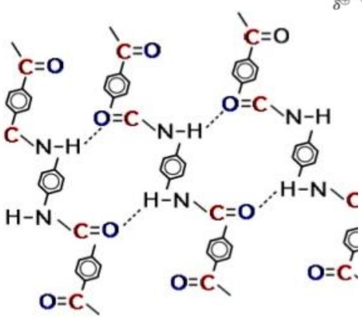
Importance of H-bonding

- H bonds are of fundamental importance in biological molecules, they help to stabilize the structure of proteins, and are responsible for the way DNA carries genetic information.
- Ice floating, fish surviving winter!



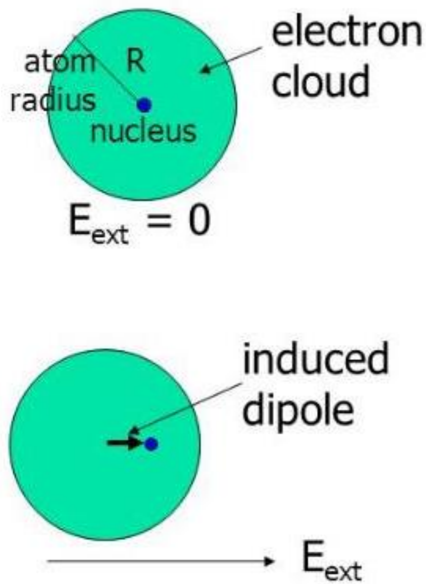


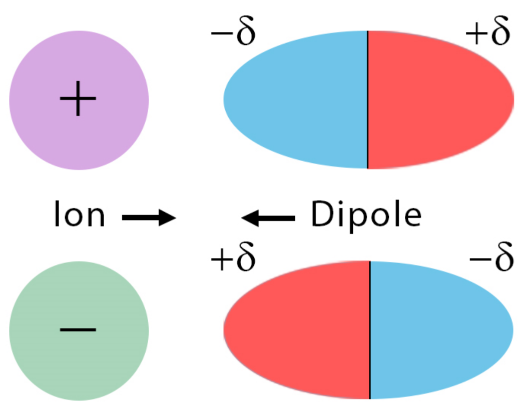




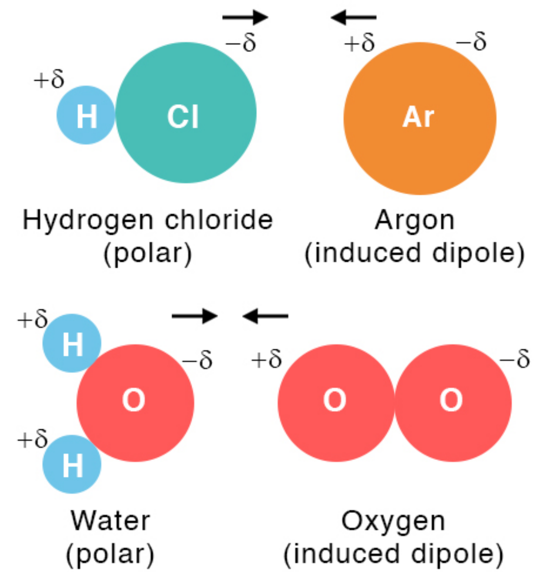
Induced dipole

- When electrically neutral atom where valence electrons are uniformly distributed around the nucleus is placed in an external electric field, E_{ext}.
- Due to attractive force on electron and repulsive force in nucleus by external electric field, distribution of valence electrons is no longer uniform.





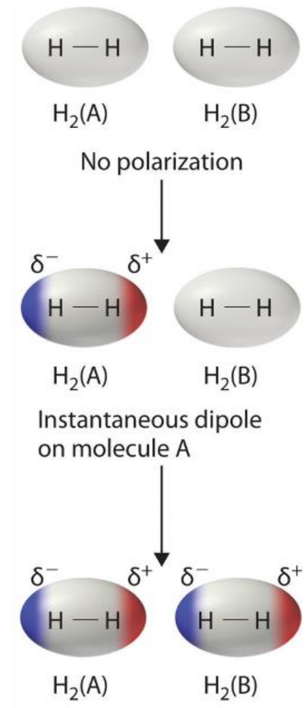
Ion-induced dipole

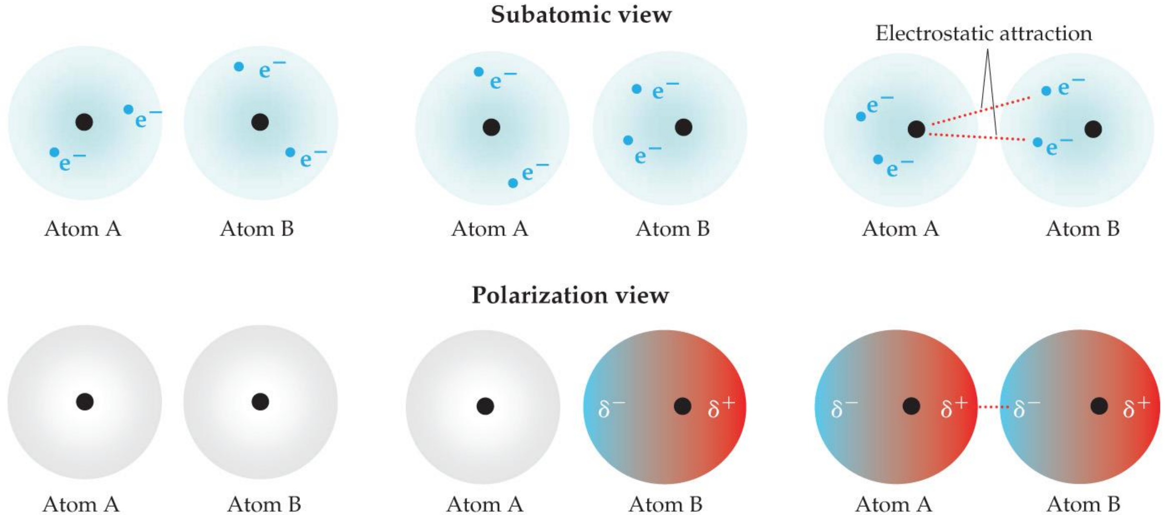


dipole-induced dipole interaction

London-Dispersion Forces

- While the electrons in the 1*s* orbital of helium would repel each other, it does happen that they occasionally wind up on the same side of the atom.
- At that instant, then, the helium atom is polar, with an excess of electrons on the left side and a shortage on the right side.
- Another helium nearby, then, would have a dipole induced in it, as the electrons on the left side of helium atom 2 repel the electrons in the cloud on helium atom 1.



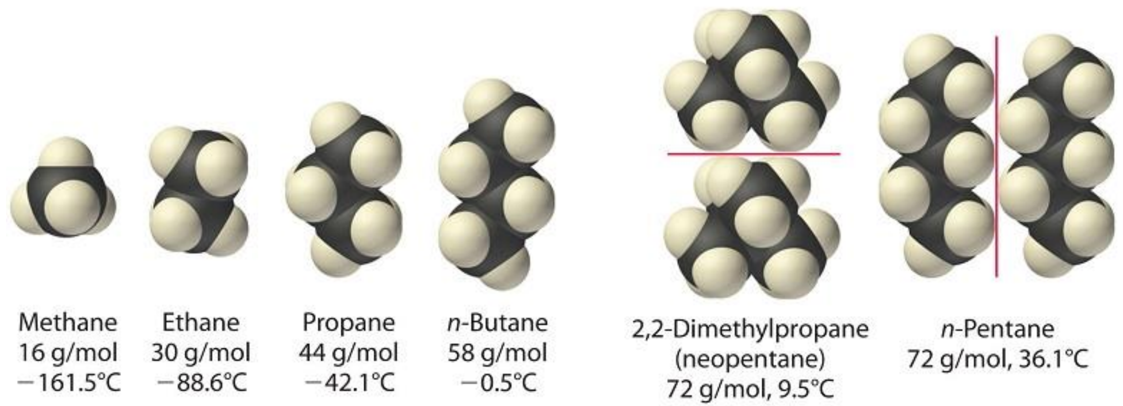


(a) Two helium atoms, no polarization **(b)** Instantaneous dipole on atom B **(c)** Induced dipole on atom A

London dispersion forces, or dispersion forces, are attractions between an instantaneous dipole and an induced dipole.

London-Dispersion Forces

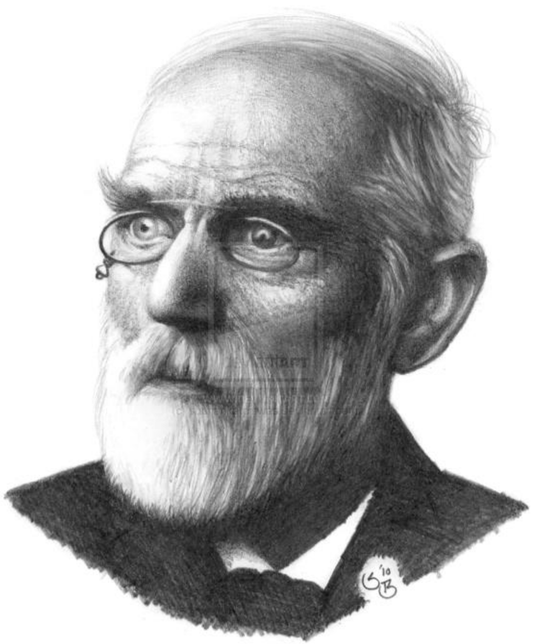
- These forces are present in all molecules, whether they are polar or nonpolar, BUT THEY ARE THE ONLY ONES THAT EXIST IN NON-POLAR MOLECULES!!!
- The tendency of an electron cloud to distort in this way is called polarizability.
- The greater the polarizability of the molecule, the more easily its electron cloud can be distorted to give a momentary dipole.
- Larger molecules tend to have greater polarizabilities. Polarizability and dispersion forces increases with molecular mass



(a) Increasing mass and boiling point

(b) Increasing surface area and boiling point

van der Waals forces



- Umbrella term for
  - London dispersion forces
  - Dipole-dipole forces
  - Dipole-induced-dipole attractions

- Refers to forces between molecules that do not involve electrostatic interactions between ions or bond formation

INTERMOLECULAR FORCES VERSUS INTRAMOLECULAR FORCES

INTERMOLECULAR FORCES	INTRAMOLECULAR FORCES
Intermolecular forces are the forces that hold molecules in a substance	Intramolecular forces are the forces that hold atoms in a molecule
Weaker than intramolecular forces	Stronger than intermolecular forces
Determine the state of matter (solid/liquid/gas) and their physical properties such as melting/boiling point etc.	Determine chemical behavior of a substance
Attractive forces	Chemical bonds
Categorized into dipole-dipole forces, London dispersion forces and hydrogen bonding forces	Categorized into covalent, ionic and metal bonds

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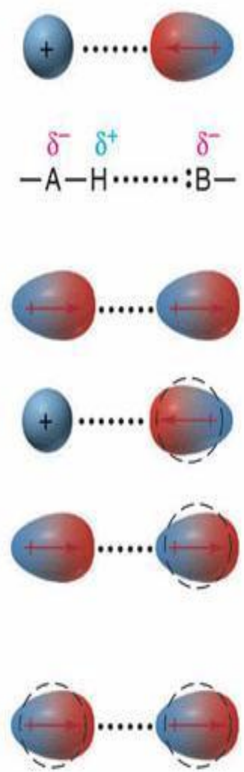
Relative strength of intermolecular forces

Nonbonding (Intermolecular)



Ion-dipole	Ion charge–dipole charge	40–600	$Na^+ \cdots OH_2$
H bond	Polar bond to H–dipole charge (high EN of N, O, F)	10–40	:Ö—H····:Ö—H (each O bonded to H)
Dipole-dipole	Dipole charges	5–25	I—Cl····I—Cl
Ion–induced dipole	Ion charge–polarizable e$^-$ cloud	3–15	$Fe^{2+} \cdots O_2$
Dipole–induced dipole	Dipole charge–polarizable e$^-$ cloud	2–10	H—Cl····Cl—Cl
Dispersion (London)	Polarizable e$^-$ clouds	0.05–40	F—F····F—F

Potential Energy Surface

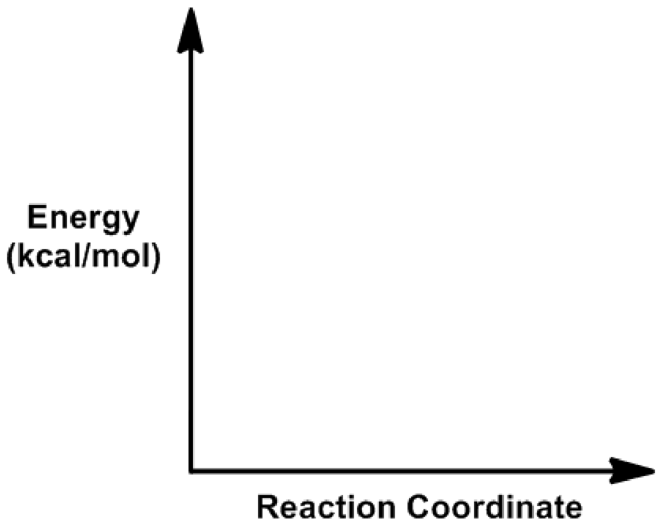
- Atoms in a molecule are held together by chemical bonds. When the atom is distorted, the bonds are stretched or compressed, in which increases the potential energy of its system.

- As the new geometry is formed, the molecule stays stationary. Therefore, the energy of the system is not caused by the kinetic energy but depending on the position of the atoms (potential).

- Energy of a molecule is a function of the position of the nuclei. When nuclei moves, electron readjusts quickly. The relationship between this molecular energy and molecular geometry (position) is mapped out with potential energy surface.

Potential Energy Surface

A **potential energy surface** (PES) represents the relationship between the energy of a molecule and its geometry.

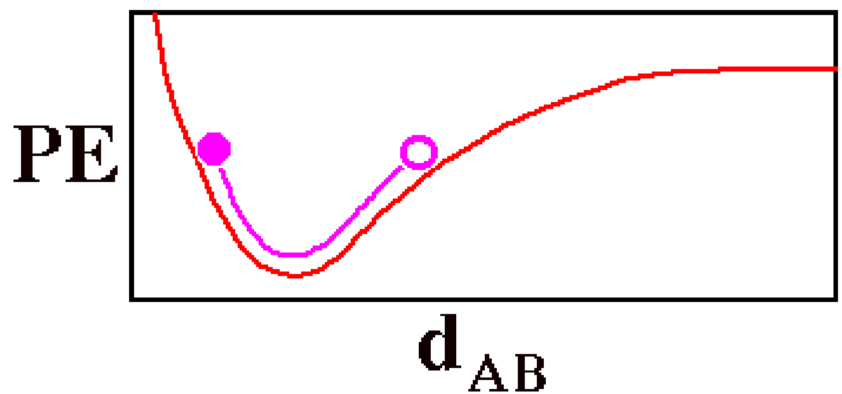


The y-axis is energy of the molecule, which relates how stable two molecules, or two conformations of the same molecule are in relation to one another. The more negative a molecule's energy, or the lower on the y-axis, the more stable it is.

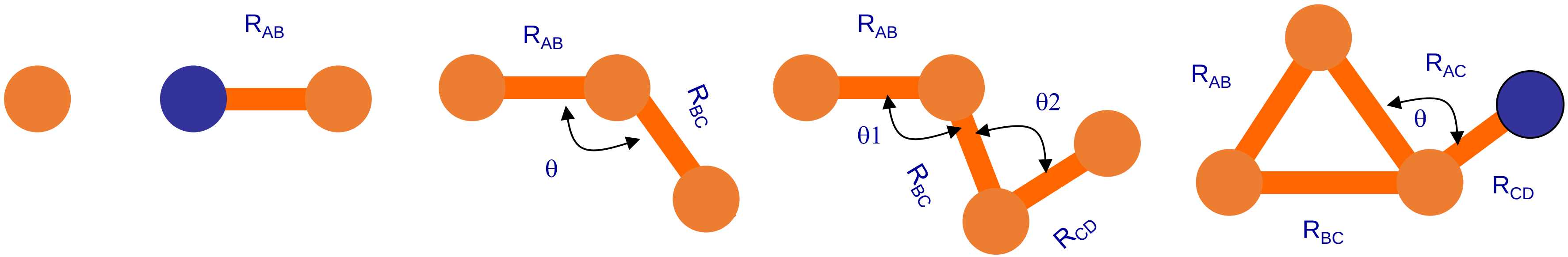
Potential Energy Surface

- **Does not change if it translated or rotated in space**
- **Depends on a molecule's internal coordinates**

- Fully specifying the location of a diatomic molecule, AB, would require specifying six numbers, the x, y, and z coordinates of each of its atoms.
- Potential energy (**PE**) of the molecule as a function of the distance between the two atoms $\mathbf{d}_{AB}$.



Molecular structure and variable



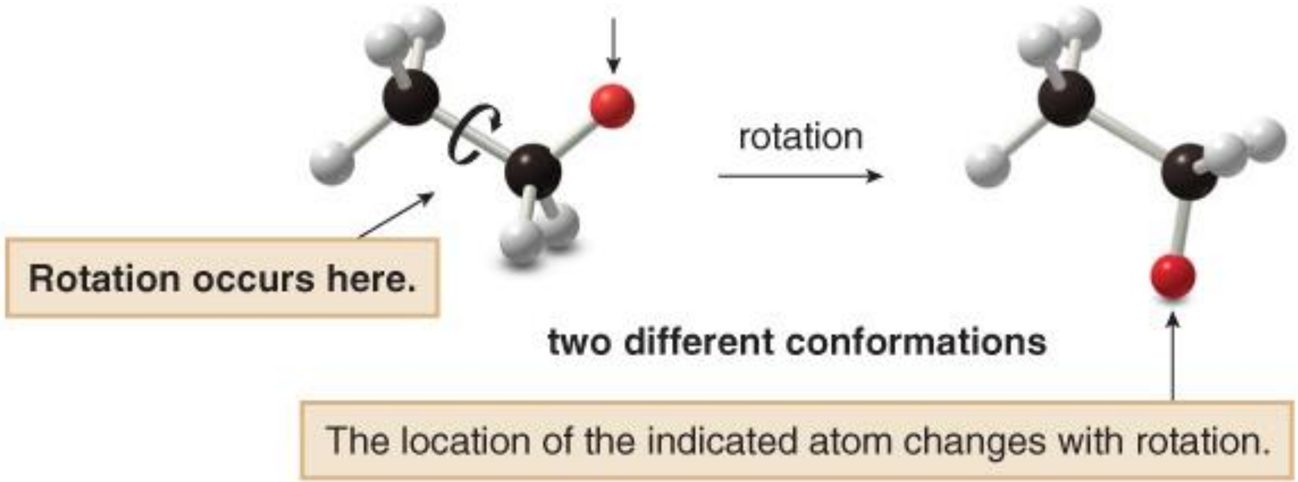
$$\text{Freedom degree} = \begin{cases} 3N-6 & \text{(Nonlinear molecules)} \\ 3N-5 & \text{(Linear molecules)} \end{cases}$$

$$\left.\begin{array}{l} N=2,\ \text{potential energy curve} \\ N \geqq 3,\ \text{potential energy hypersurface} \end{array}\right\} \text{Potential energy surfaces}$$

$$U(q_1, q_2, q_3, \ldots \ldots q_n)$$

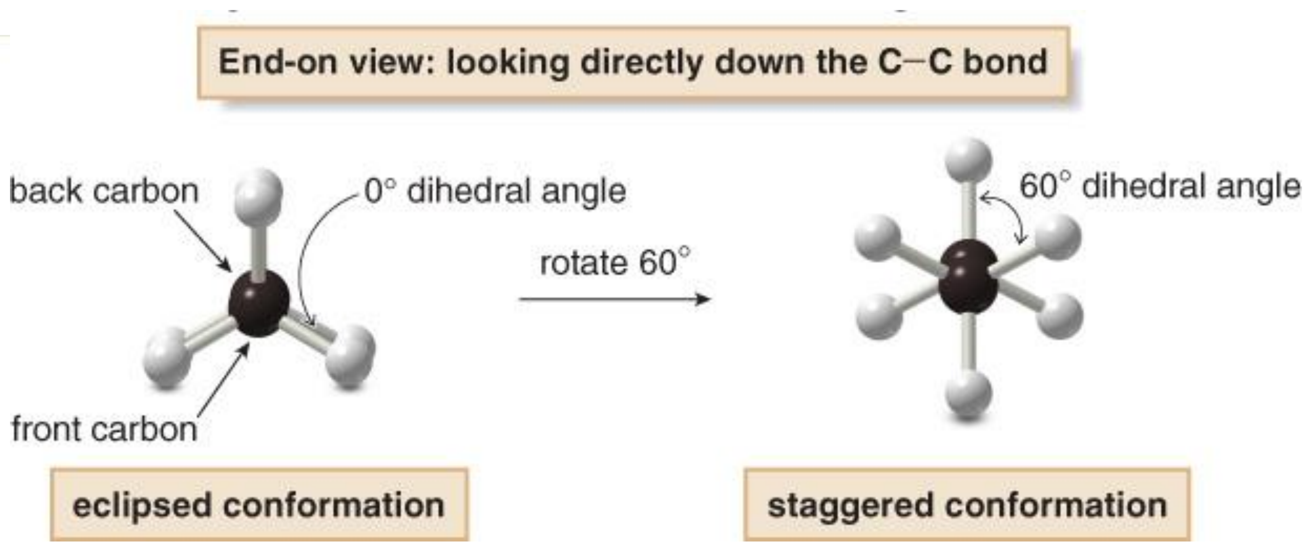
Conformations

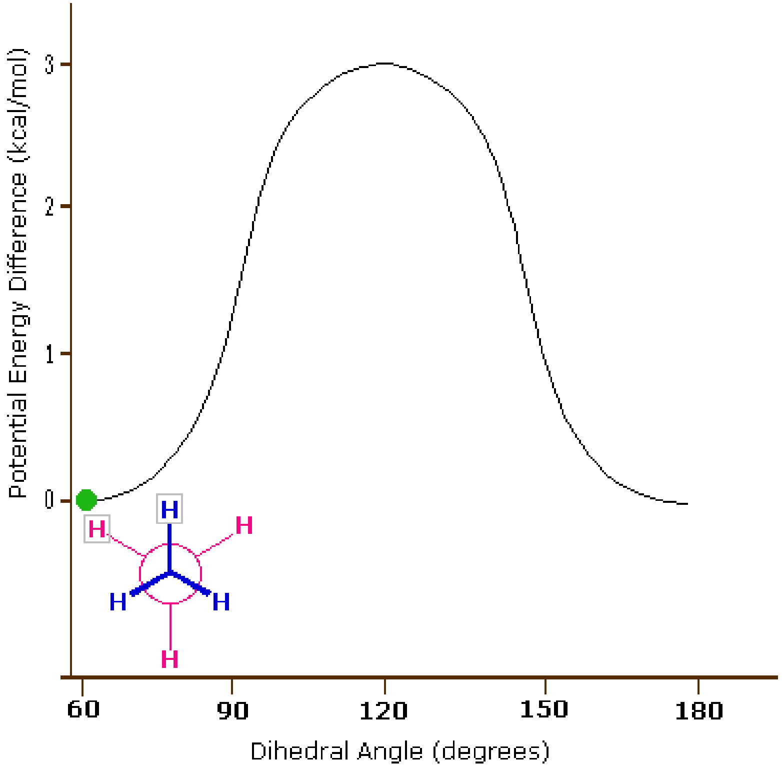
Conformations are different arrangements of atoms that are interconverted by rotation about single bonds. A particular conformation is called a conformer.

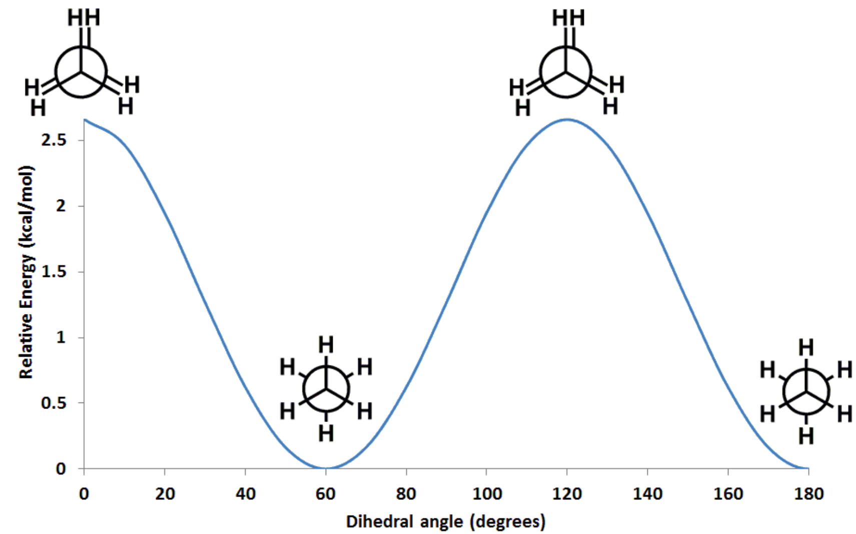


Staggered & Eclipsed Conformations of Ethane

- In the eclipsed conformation, the C—H bonds on one carbon are directly aligned with the C—H bonds on the adjacent carbon.
- In the staggered conformation, the C—H bonds on one carbon bisect the H—C—H bond angle on the adjacent carbon.



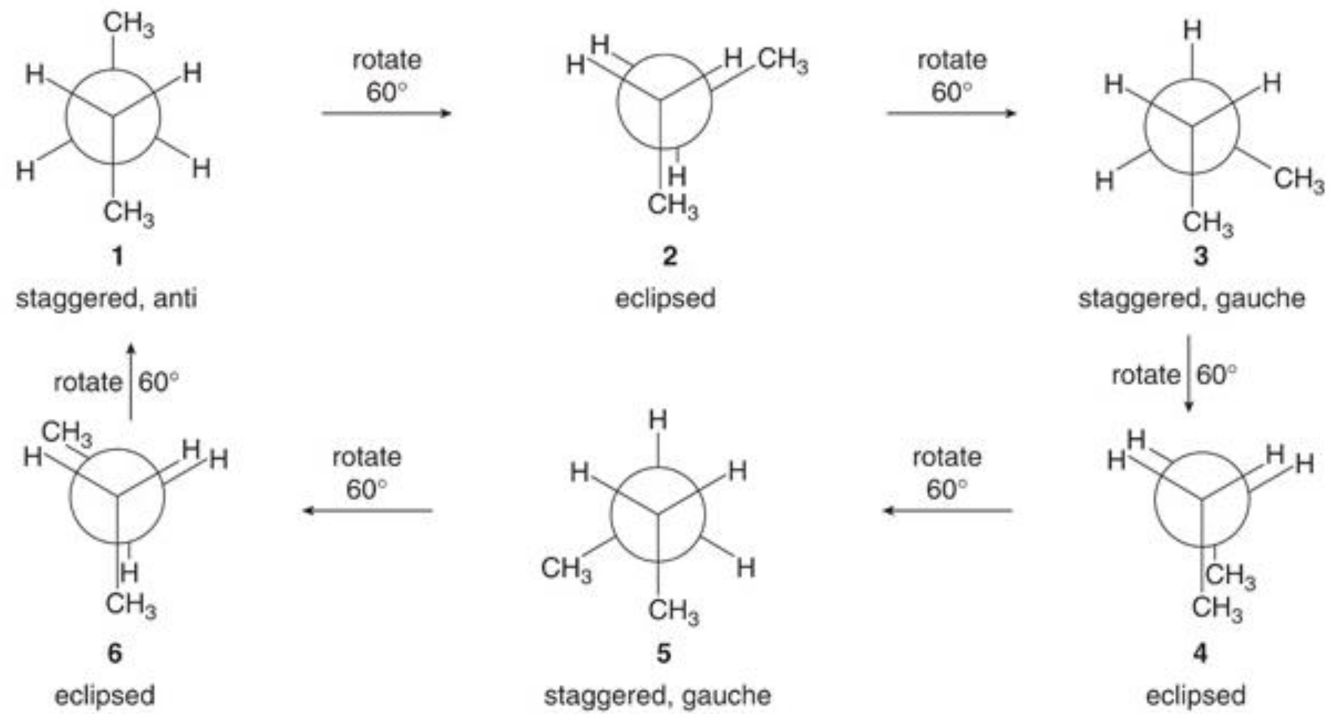


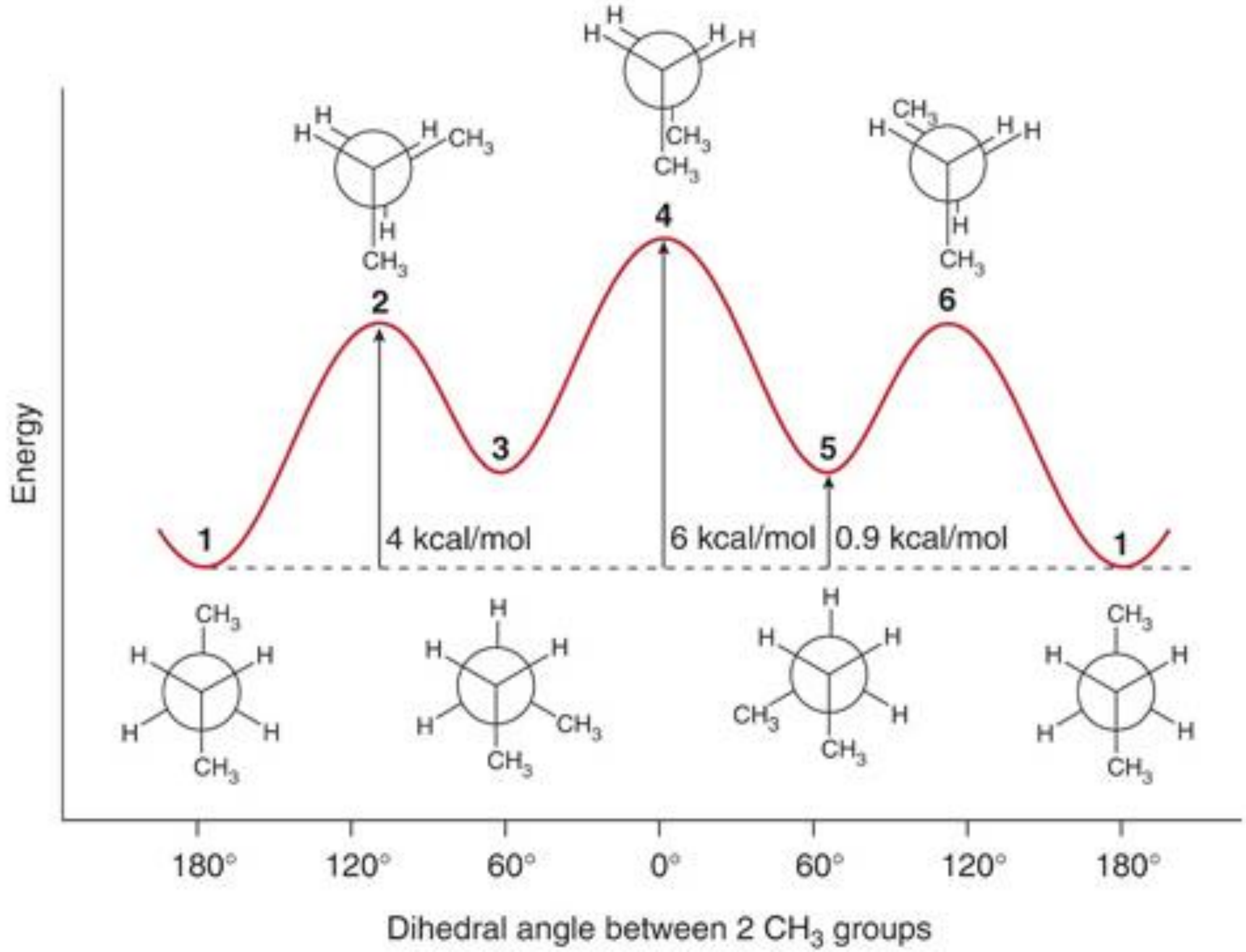


This PES shows the energy differences between the two conformations of ethane. The eclipsed conformation with the hydrogen atoms overlapping one another is about 2.6 kcal/mol higher in energy than the staggered conformation.

Conformations of Butane

- An energy minimum and maximum occur every 60^0 as the conformation changes from staggered to eclipsed. Conformations that are neither staggered nor eclipsed are intermediate in energy.





Energy
4 kcal/mol
6 kcal/mol
0.9 kcal/mol
1
2
3
4
5
6
1
180°
120°
60°
0°
60°
120°
180°
Dihedral angle between 2 CH_3 groups

Characterizing Potential Energy Surface

The most interesting points on PES's are the stationary points, where the gradients with respect to all internal coordinates are zero.

1. Minima: correspond to stable or quasi-stable species; i.e., reactants, products, intermediates.

2. Transition states: saddle points which are minima in all dimensions but one; a maximum in that dimension.

3. Higher-order saddle points: a minimum in all dimensions but n, where n > 1; maximum in the other n dimensions.

Characterizing Potential Energy Surface

$$\left(\frac{df}{dx}\right)_y = 0, \left(\frac{df}{dy}\right)_x = 0$$

At the minima and saddle point, the slope of the function is zero

$$\left(\frac{d^2f}{dx^2}\right)_y => 0$$

At the minima, the second derivatives are positive ,

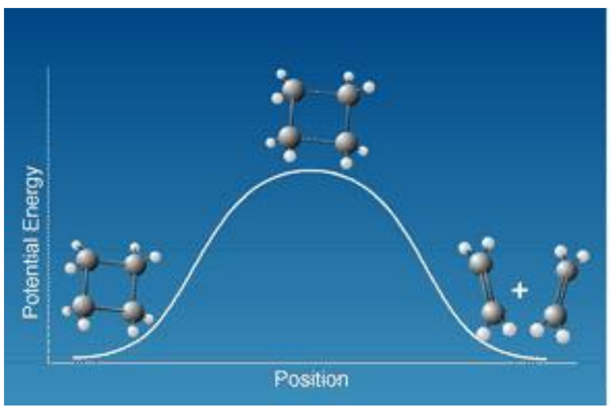
$$\left(\frac{d^2f}{dy^2}\right)_x => 0$$

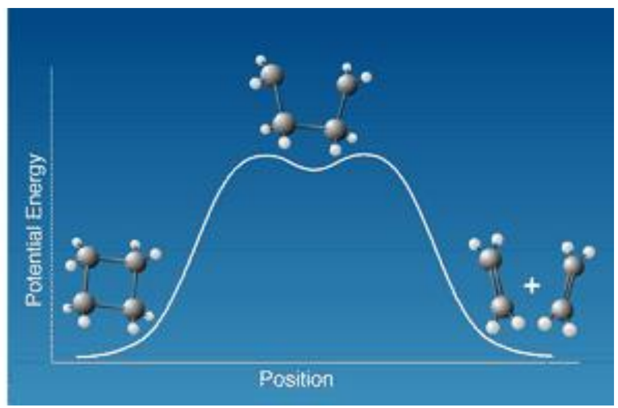
$$\left(\frac{d^2f}{dx^2}\right)_y =< 0$$

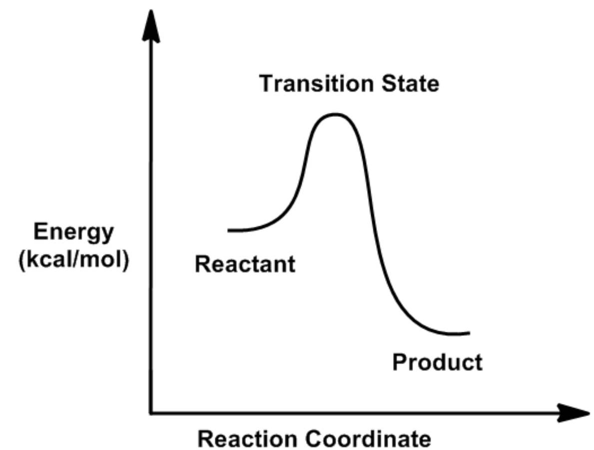
At the saddle point, the second derivatives are negative

Potential Energy Surface

- The reactant and the product are both **minima** on the PES. This indicates that they are conformations that the molecule can exist in.
- The peak in between is the **transition state** that leads from the reactant to the product.
- As the transition state is NOT a minima, the molecule can never exist in that conformation, it is not a stable conformation.







Potential Energy Functions

A potential energy surface is a mathematical function that gives the energy of a molecule as a function of its geometry.

- Quantum Mechanics provides an energy function which can be exact in principle and works for any molecule.

- Molecular Mechanics provides this energy as a function of stretches, bends, torsions, etc. This is an approximate model that breaks down in some situations (e.g., breaking bonds).

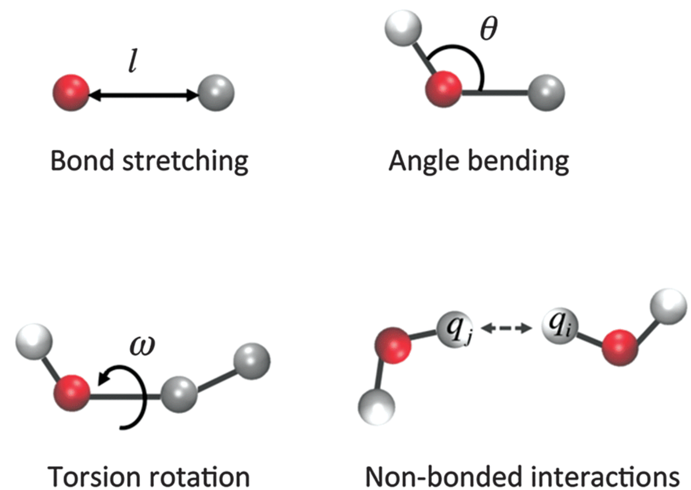
Potential Energy Functions

$$E = E_{covalent} + E_{noncovalent}$$

can be further expanded to:

$$E_{covalent} = E_{bond} + E_{angle} + E_{dihedral}$$

$$E_{noncovalent} = E_{electrostatic} + E_{vdw}$$

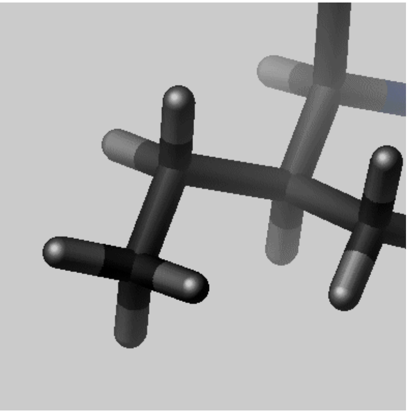


Bond Length Potentials

- Whenever a bond is compressed or stretched the energy goes up.
- The energy potential for bond stretching and compressing is described by an equation similar to Hooke's law for a spring.

$$V_b\left(r_{ij}\right) = \frac{1}{2}k^b_{ij}(r_{ij} - b_{ij})^2$$

Sum over all bonds in the structure



l_o – expected/natural bond length
k_l – force constant
l – actual/observed bond length

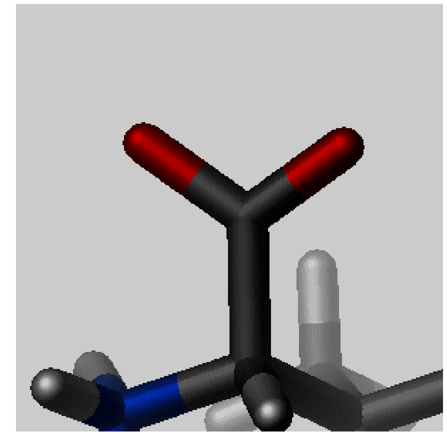
This is the approximation to the energy of a bond as a function of displacement from the ideal bond length, b_0. The force constant, K_b, determines the strength of the bond. Both ideal bond lengths b_0 and force constants K_b are specific for each pair of bound atoms, i.e. depend on chemical type of atoms-constituents.

Bond Angle Potentials

As the bond angle is bent from the norm, the energy goes up.

$$V_a(\theta_{ijk}) = \frac{1}{2}k^{\theta}_{ijk}(\theta_{ijk} - \theta^{0}_{ijk})^2$$

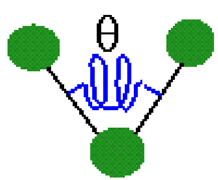
Sum over all angles in the structure

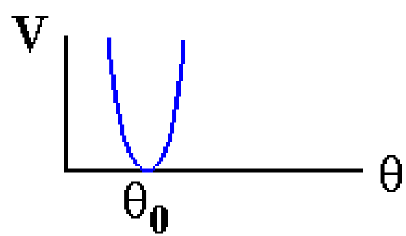


ϕ_o – expected/natural bond angle
k_ϕ – force constant
ϕ – actual/observed bond angle

It is the alteration of bond angles theta from ideal values q_0 , which is also represented by a harmonic potential. Values of q_0 and K_q depend on chemical type of atoms constituting the angle.

Angles





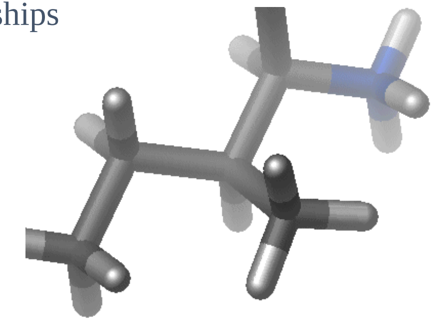
Plot of Potential Energy Function for Bond Length

Dihedral Angle Potentials

- As the dihedral angle is bent from the norm, the energy goes up.
- The torsion potential is a Fourier series that accounts for all 1-4 through-bond relationships

$$\sum_{torsions} A_n \left[1 + \cos(n\phi - \phi_0)\right]$$

Sum over all dihedrals in the structure



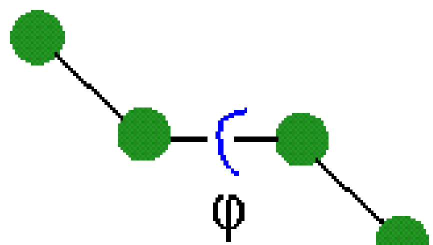
…

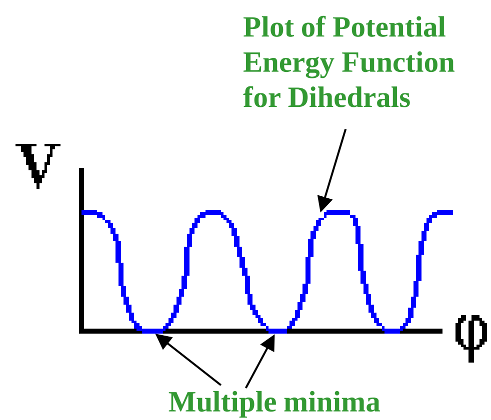
- ϕ_o – expected dihedral
- A_n – force constant for each term

ϕ – actual/observed dihedral

- n – multiplicity

Torsions





Improper Dihedral Angle Potentials

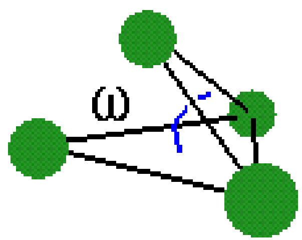
- As the improper dihedral is bent from the norm, the energy goes up.

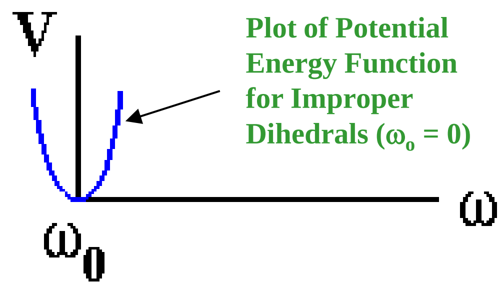
$$\sum_{impropers} k_{\omega}(\omega - \omega_0)^2$$

Sum over all improper dihedrals in the structure

- ω_o – expected improper dihedral (usually set to 0°)
- k_{ω} – force constant
- ω – actual/observed improper dihedral

Improper Dihedrals

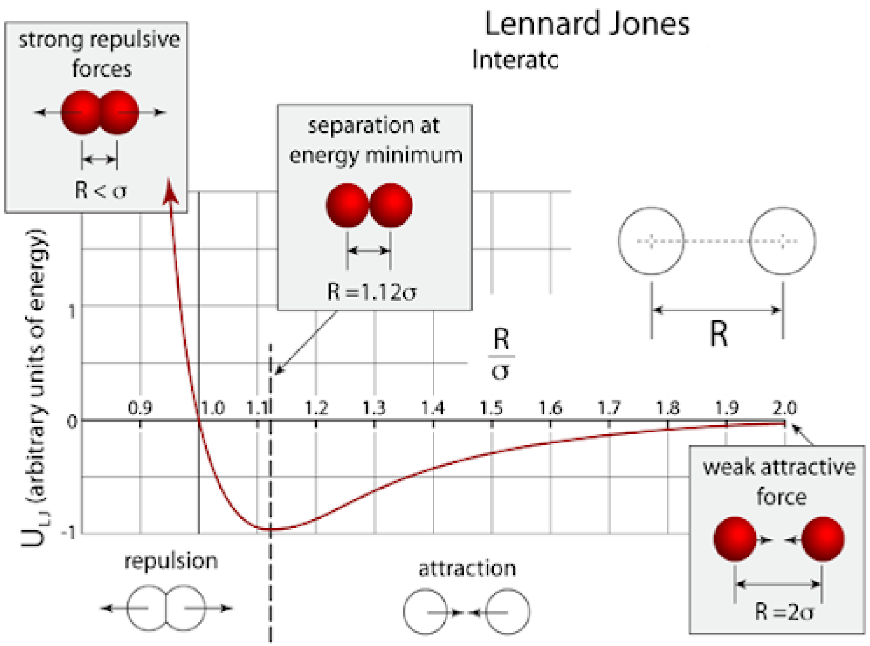




Plot of Potential Energy Function for Improper Dihedrals (ω_o = 0)

Non-bonding: van der Waals Potential

One of the most widely used functions for the van der Waals potential is the Lennard-Jones. It is a compromise between accuracy and computability.



$$E_{Lennard-Jones} = \sum_{\substack{nonbonded\\pairs}} \left(\frac{A_{ik}}{r_{ik}^{12}} - \frac{C_{ik}}{r_{ik}^{6}} \right)$$

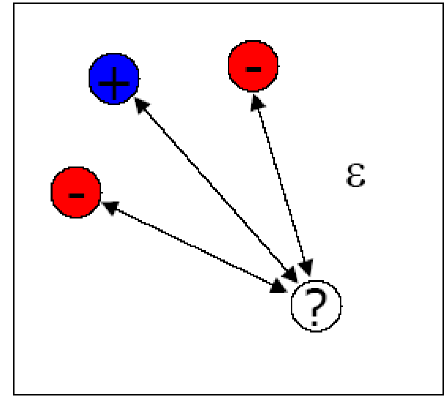
The Constants A and C depend on the atom types, and are derived from experimental data.

It can also be written as:

$$U = \sum_{i<j} \sum 4\varepsilon_{ij} \left[\left(\frac{\sigma_{ij}}{r_{ij}} \right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}} \right)^{6} \right]$$

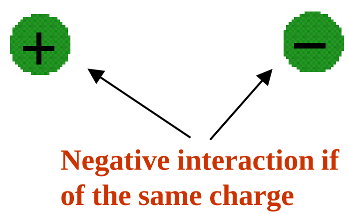
Non-bonding: Electrostatic Interactions

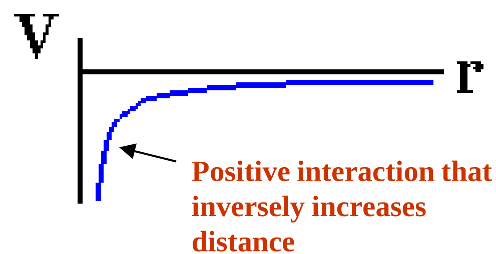
- electrostatic interaction
  - Electrostatic interaction of charged atoms
  - Long-range forces
  - Coulomb's Law



$$E_{pot} = \sum_{i} \frac{q_i q_j}{4 \cdot \pi \cdot \varepsilon \cdot r_{ij}}$$

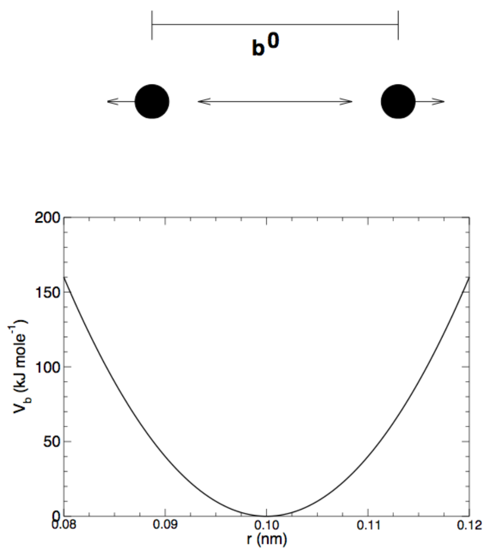
Electrostatics





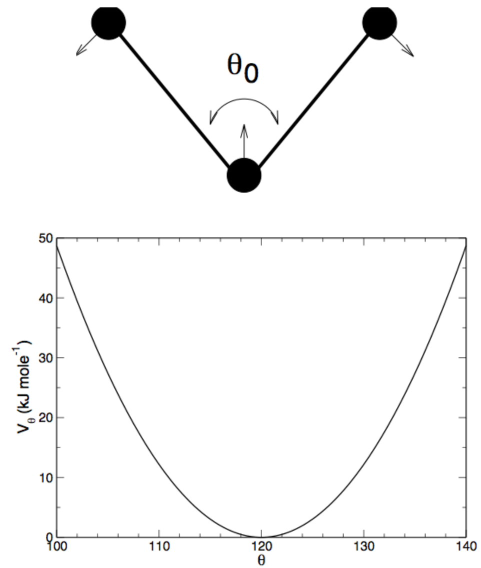
Potential Energy Functions

$$E_{covalent} = E_{bond} + E_{angle} + E_{dihedral}$$



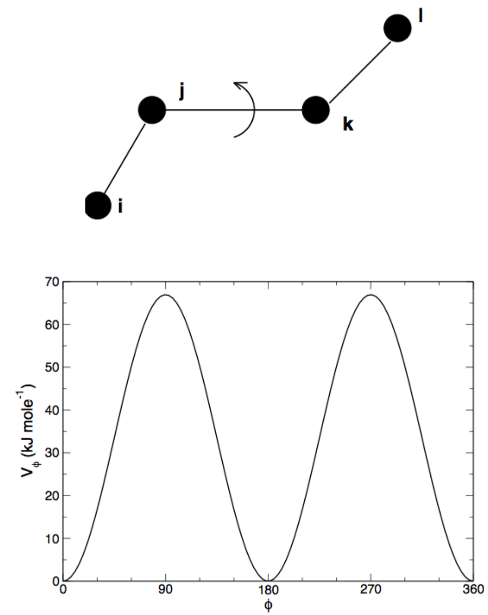
$$V_b\,(r_{ij}) = \frac{1}{2}k^b_{ij}(r_{ij} - b_{ij})^2$$

Bond Stretching Potential



$$V_a(\theta_{ijk}) = \frac{1}{2}k^{\theta}_{ijk}(\theta_{ijk} - \theta^0_{ijk})^2$$

Harmonic Angle Potential

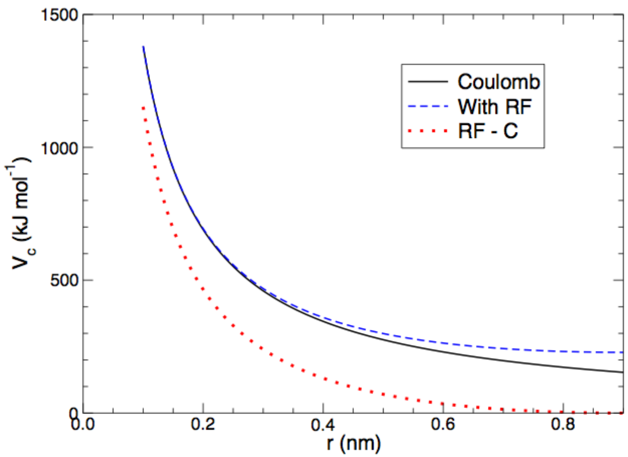


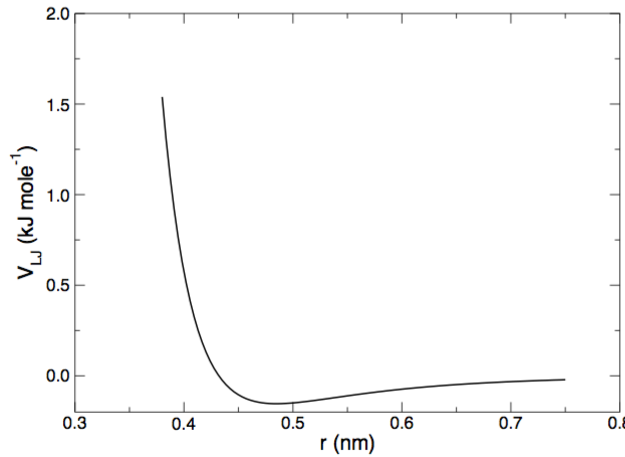
$$V_d(\phi_{ijkl}) = k_\phi(1 + \cos(n\phi - \phi_s))$$

Dihedral Potential

Potential Energy Functions

$$E_{noncovalent} = E_{electrostatic} + E_{vdw}$$





$$V_c(r_{ij}) = f \frac{q_i q_j}{\varepsilon_r r_{ij}}$$

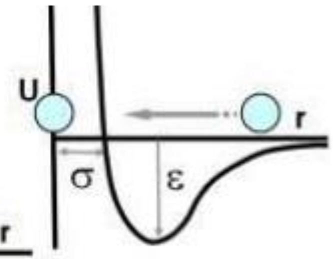
$$V_{LJ}(r_{ij}) = \sum_{\substack{nonbonded \\ pairs}} \left(\frac{A_{ik}}{r_{ik}^{12}} - \frac{C_{ik}}{r_{ik}^{6}} \right)$$

Coulombic Potential

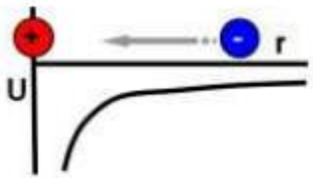
Lennard-Jones Potential

Example: Potential Energy Function

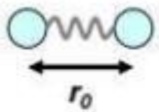
$$U = \sum_{i<j} \sum 4\varepsilon_{ij}\left[\left(\frac{\sigma_{ij}}{r_{ij}}\right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}}\right)^{6}\right]$$



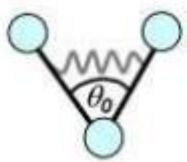
$$+ \sum_{i<j} \sum \frac{q_i q_j}{4\pi\varepsilon_0 r_{ij}}$$



$$+ \sum_{bonds} \frac{1}{2} k_b (r - r_0)^2$$



$$+ \sum_{angles} \frac{1}{2} k_a (\theta - \theta_0)^2$$



$$+ \sum_{torsions} k_\phi \left[1 + \cos(n\phi - \delta)\right]$$